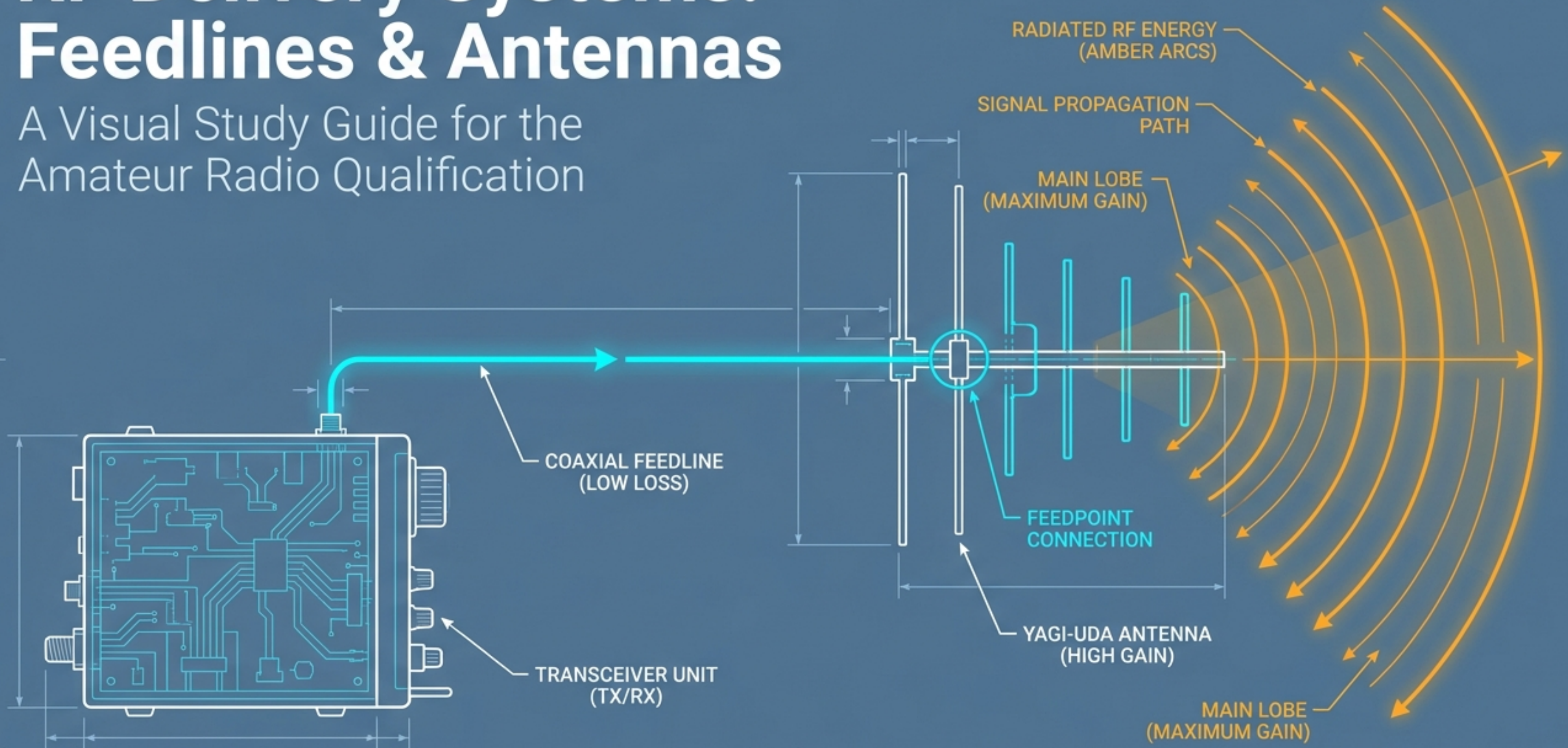
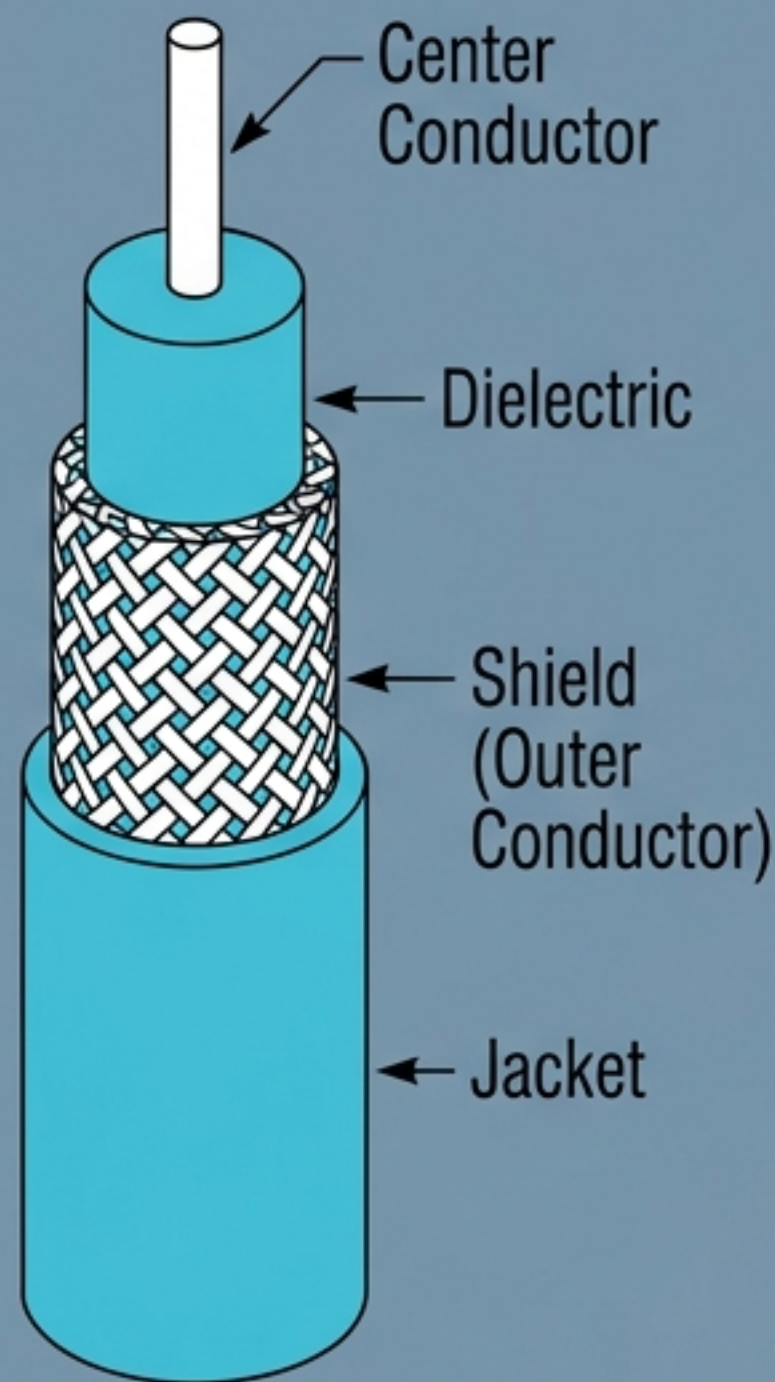


RF Delivery Systems: Feedlines & Antennas

A Visual Study Guide for the
Amateur Radio Qualification



FEEDLINE FUNDAMENTALS: COAXIAL VS. OPEN-WIRE



COAXIAL CABLE (UNBALANCED)

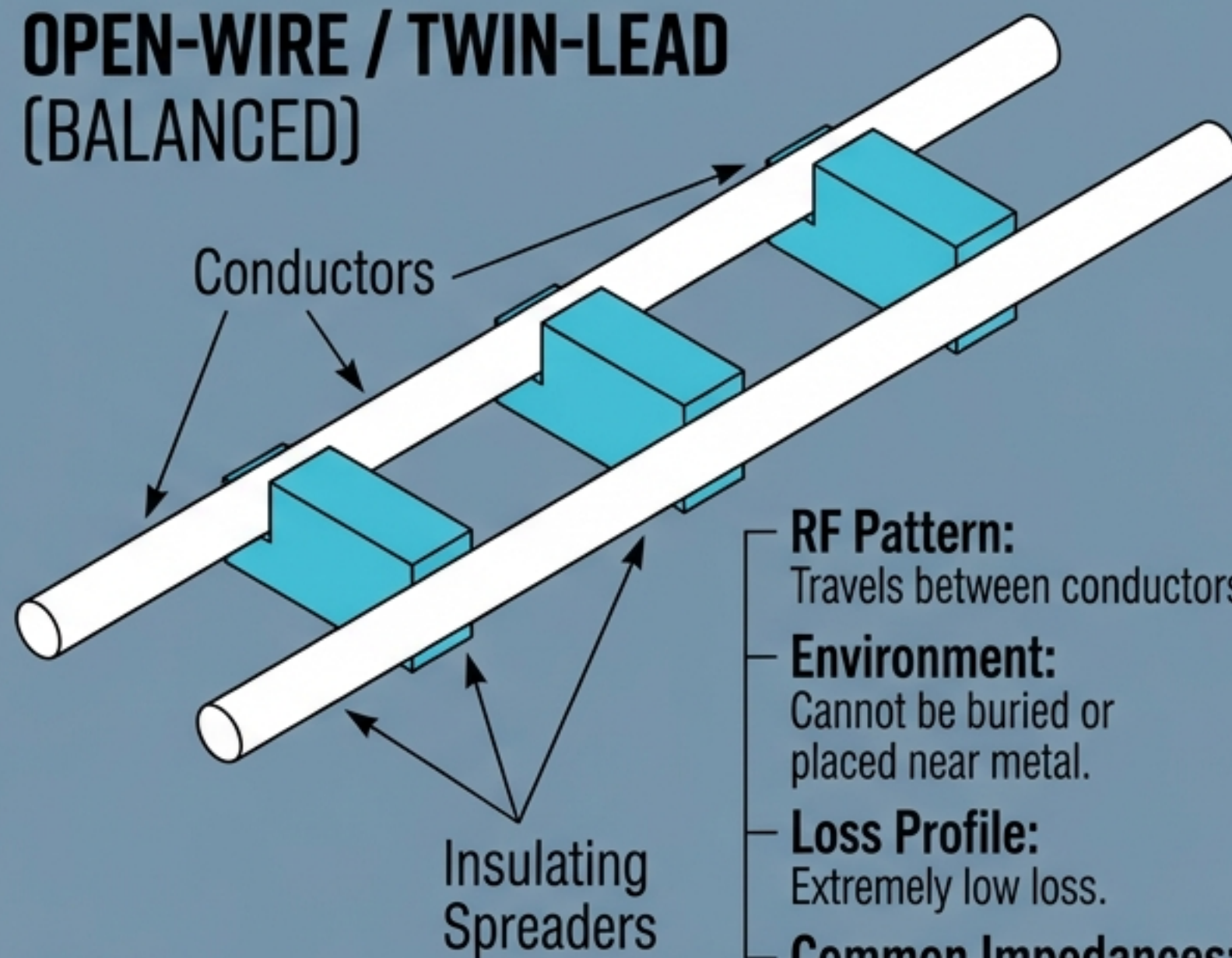
RF Pattern:
Travels inside the shield.

Environment:
Immune to nearby metal;
can be buried.

Loss Profile:
Higher loss at VHF/UHF.

Common Impedances:
50 ohms (RG-8, RG-58,
RG-213) and 75 ohms
(RG-59).

OPEN-WIRE / TWIN-LEAD (BALANCED)

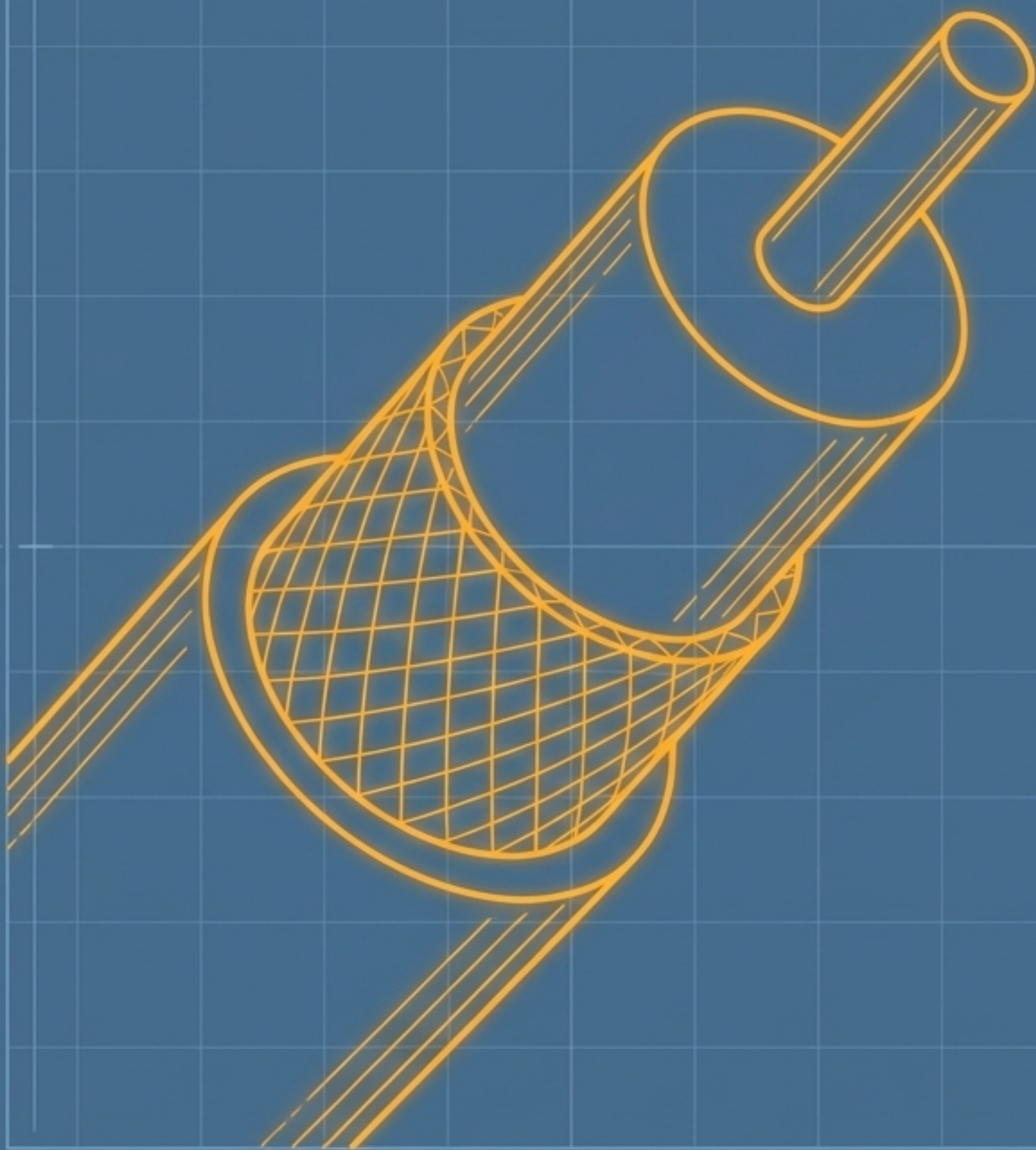


RF Pattern:
Travels between conductors.

Environment:
Cannot be buried or
placed near metal.

Loss Profile:
Extremely low loss.

Common Impedances:
300 ohms, 450 ohms,
600 ohms.

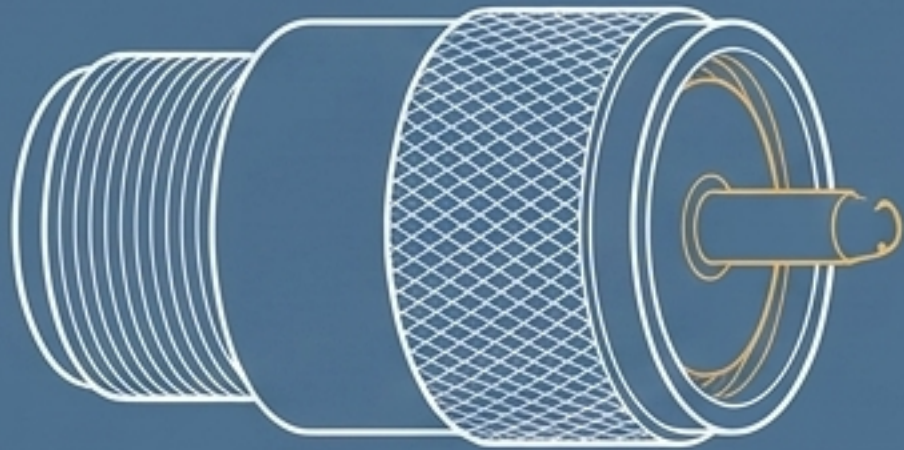


Diagnostic Check: Transmission Lines

Why do most amateur radio antenna systems use coaxial cable, rather than other types of transmission line?

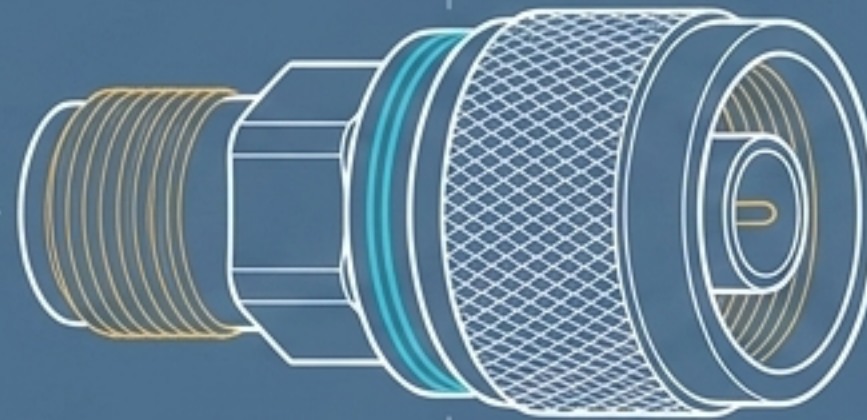
- A) Less loss
- B) More power handling capability
- C) Less expensive
- D) More usable in a wide variety of settings

Connectors & The Balun Transition



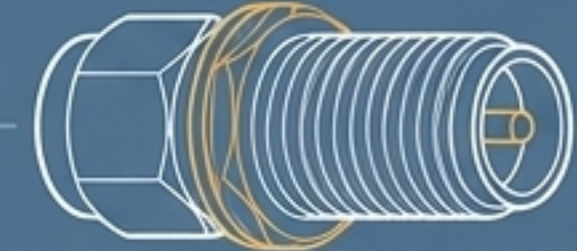
PL-259

Rugged, standard HF/VHF



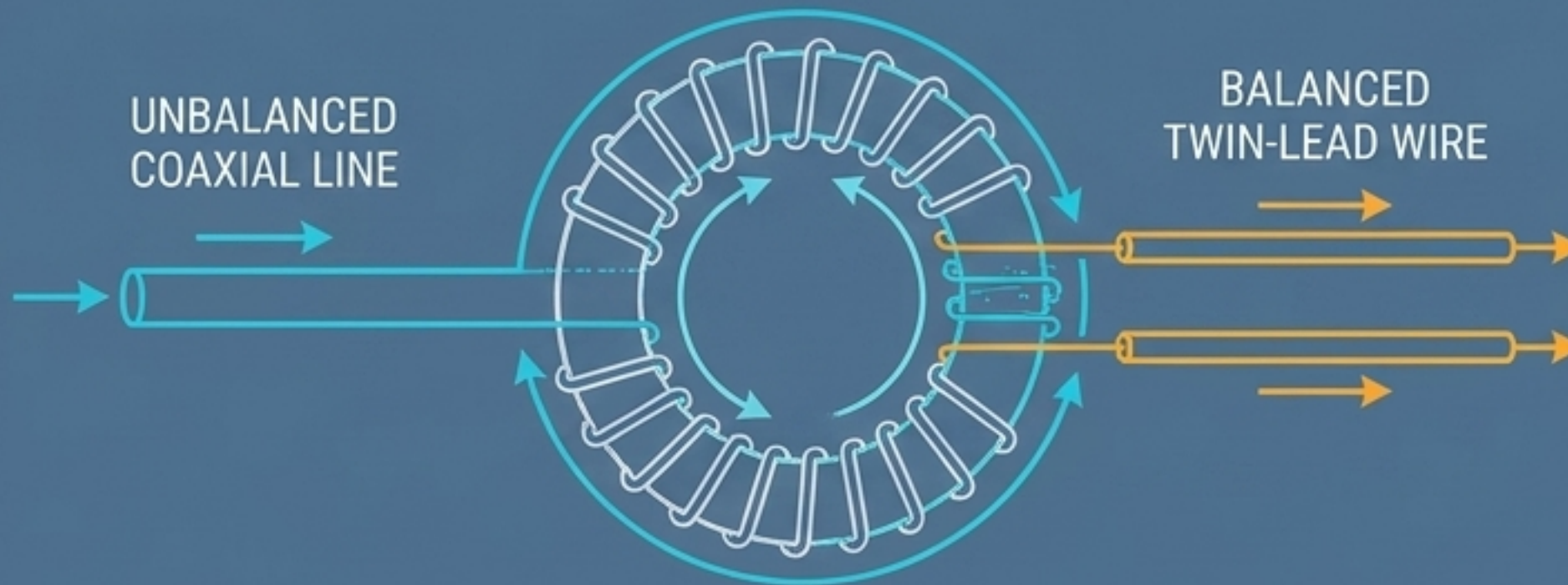
N-Type

Moisture resistant, ideal for UHF



SMA

Miniature, low power handhelds



The Balun (Balanced-to-Unbalanced)

- Function 1: Prevents the coaxial shield from radiating RF energy.
- Function 2: Matches impedance between disparate systems.

Example: A 4:1 ratio balun converts a 300 ohm antenna to match a 75 ohm coaxial line.



Diagnostic Check: Feedpoint Components

What device can be installed to feed a balanced antenna with an unbalanced transmission line?

- A) A balun
- B) A triaxial transformer
- C) A wave trap
- D) A loading coil

Impedance Matching: SWR & Tuners



Forward vs. Reflected Power

Forward: Energy moving toward the load.
Reflected: Energy bouncing back due to mismatched impedance.

SWR (Standing Wave Ratio)

Perfect Match = 1:1
>1.5:1 causes transmitter to automatically reduce power.
Calculated as Load Impedance / Line Impedance.

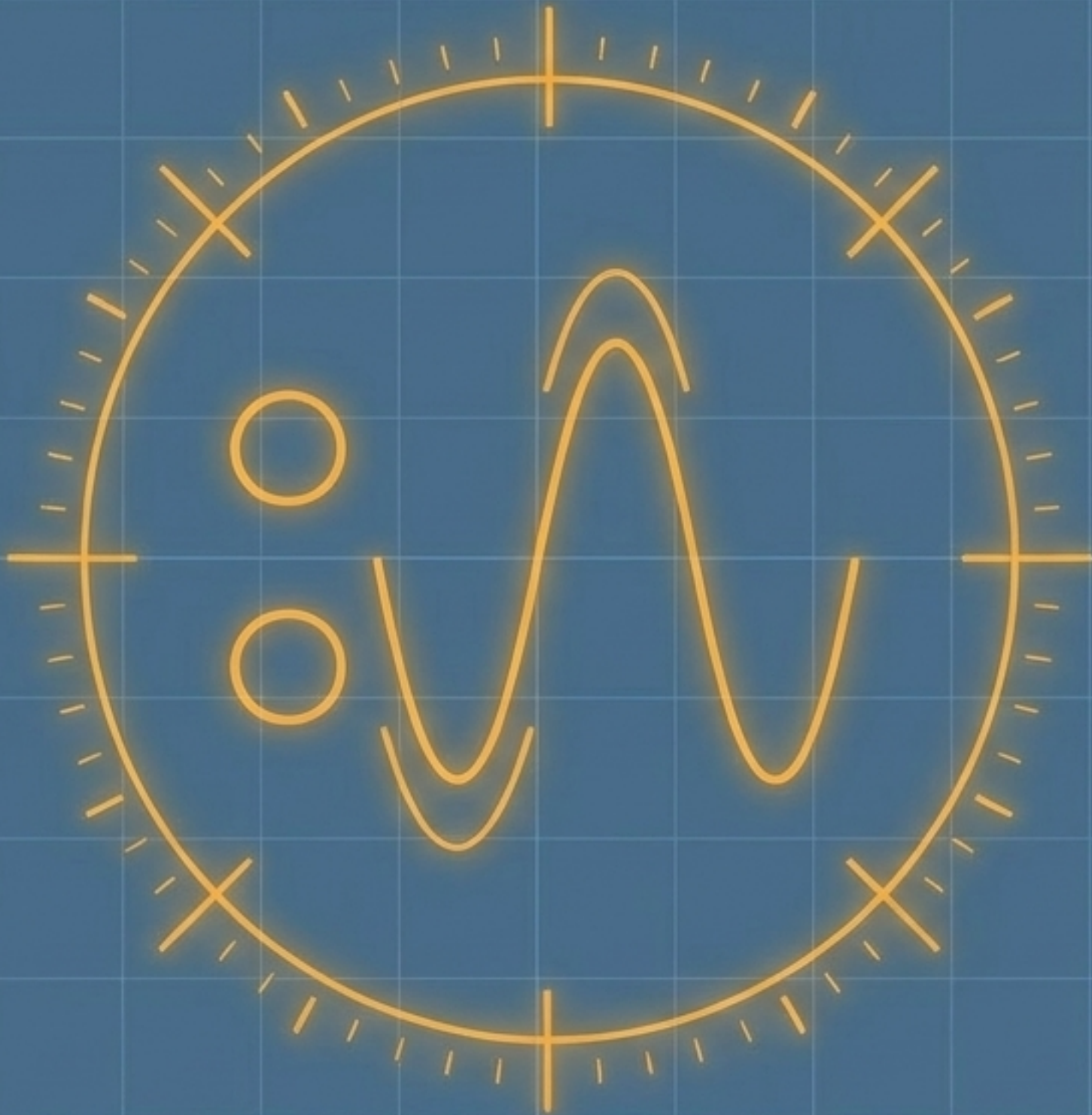
Antenna Tuner

Cancels unwanted reactance to present a perfect 50 ohm load to the transmitter.

Diagnostic Check: SWR Calculations

If a transmission line with a characteristic impedance of 50 ohms feeds a folded dipole with a feed point impedance close to 300 ohms, what impedance transformation ratio is needed to match the two?

- A) 9:1
- B) 6:1
- C) 2:1
- D) 4:1



Wave Fundamentals & The Radiators

Tutor's Toolkit

Free Space Wavelength:
 $L = 300 / F(\text{MHz})$

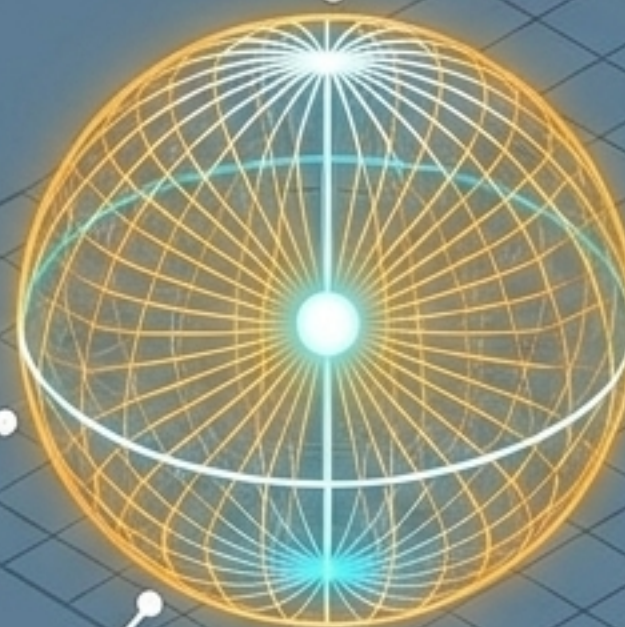
Polarization:

Determined by electric lines of force.
Parallel to earth = Horizontal.
Perpendicular = Vertical.

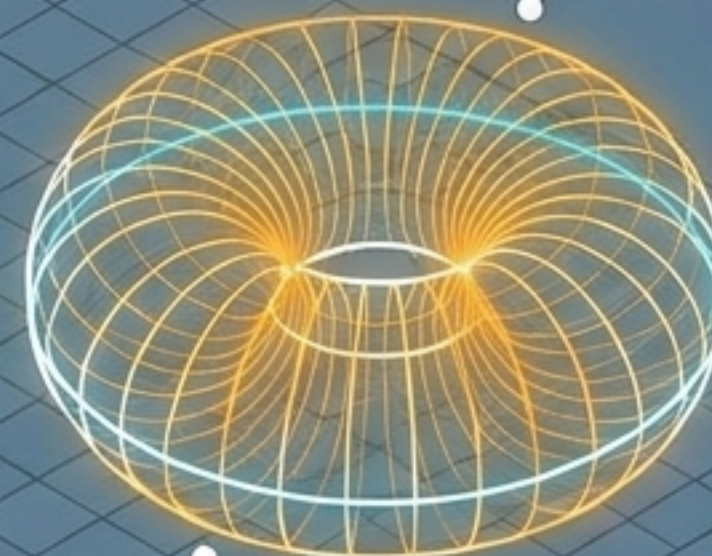
Isotropic Radiator:

A theoretical point source radiating equally in all directions (0 dBi).
Represented by a perfect sphere pattern.

Isotropic Radiator



Dipole Radiator



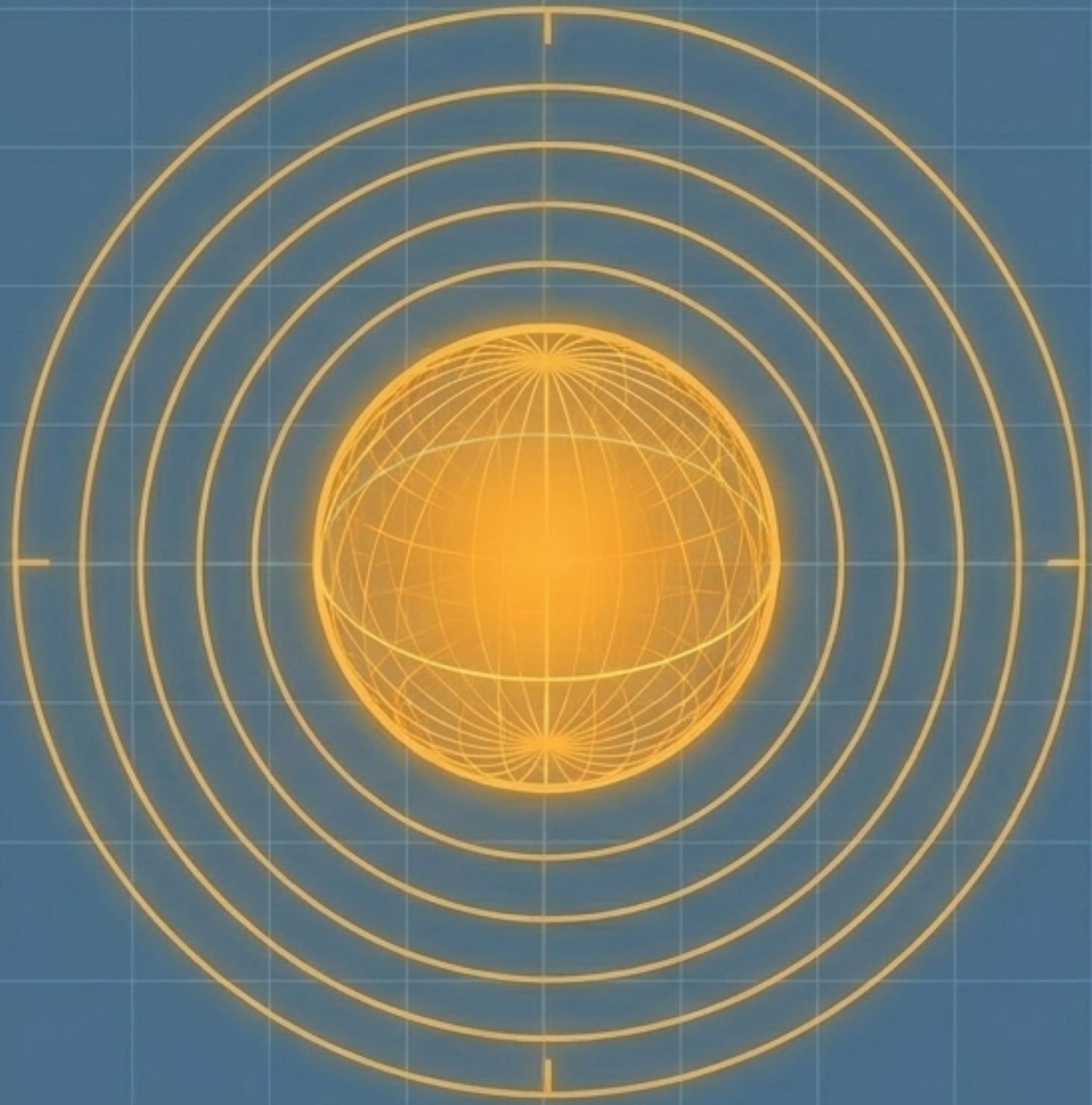
Dipole Radiator:

The real baseline antenna.
Radiates a torus (donut) pattern with 2.14 dBi gain.

Diagnostic Check: Antenna Theory

What is the three-dimensional radiation pattern of an isotropic radiator?

- A) A hemisphere
- B) A sphere
- C) A torus (donut shape)
- D) A parabola

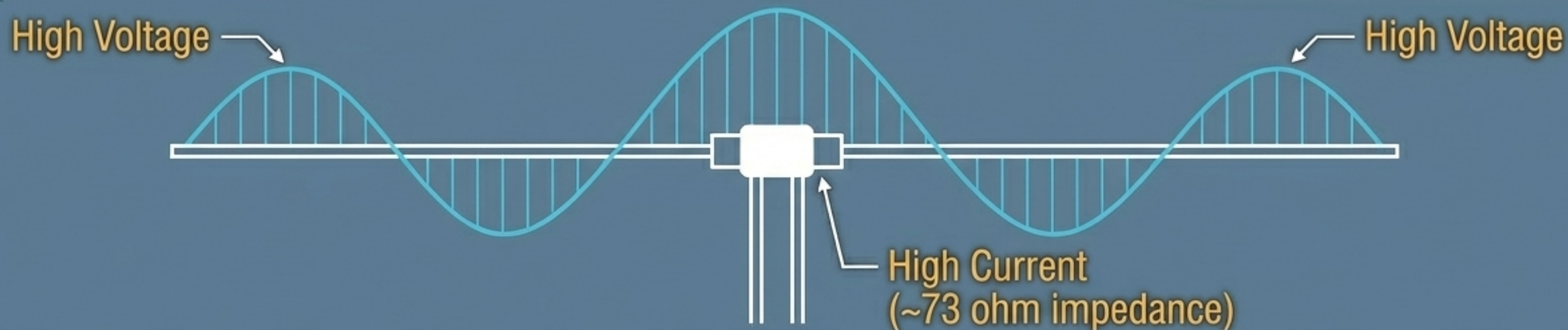


Simple Wire Antennas: The Half-Wave Dipole

Tutor's Toolkit

Electrical Length:

$$L = 142 / F(\text{MHz})$$



Inverted-V



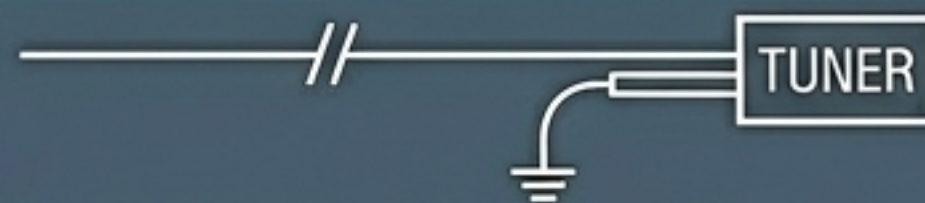
Center fed, ends droop toward ground. Requires only one tall central support.

Trap Dipole

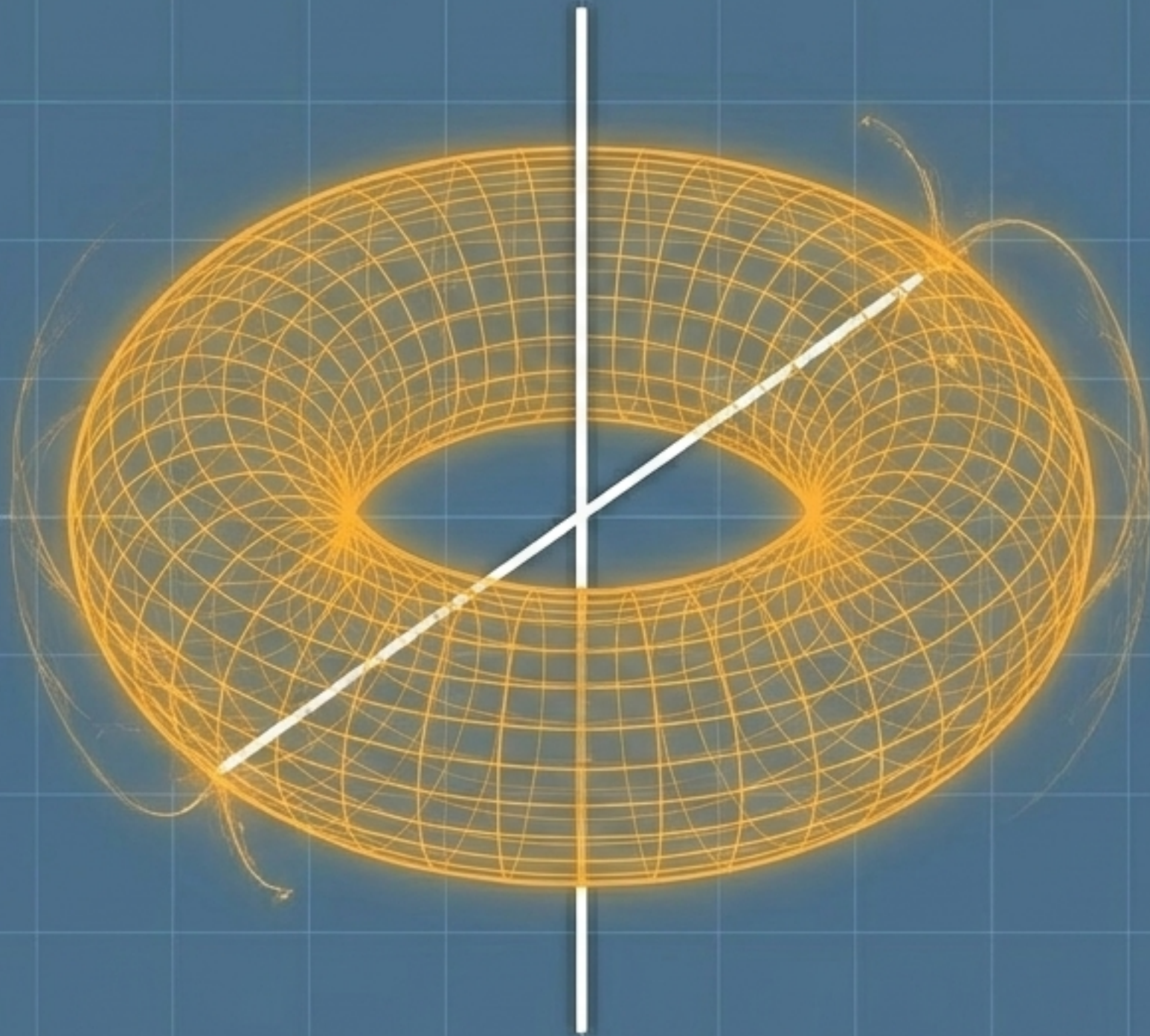


Uses LC circuits (traps) to physically isolate sections, allowing multi-band operation.

Random Wire



End-fed wire of any length. Requires an external antenna tuner to match impedance.



Diagnostic Check: Wire Antennas

What is the three-dimensional radiation pattern of a half-wavelength dipole in free space?

- A) Major lobes at 45-degree angles to the antenna
- B) A torus (donut shape) around the antenna
- C) A major lobe off each end of the antenna
- D) Spherical (equal radiation in all directions)

Vertical Antennas & Ground Planes

1/4-Wave Vertical •

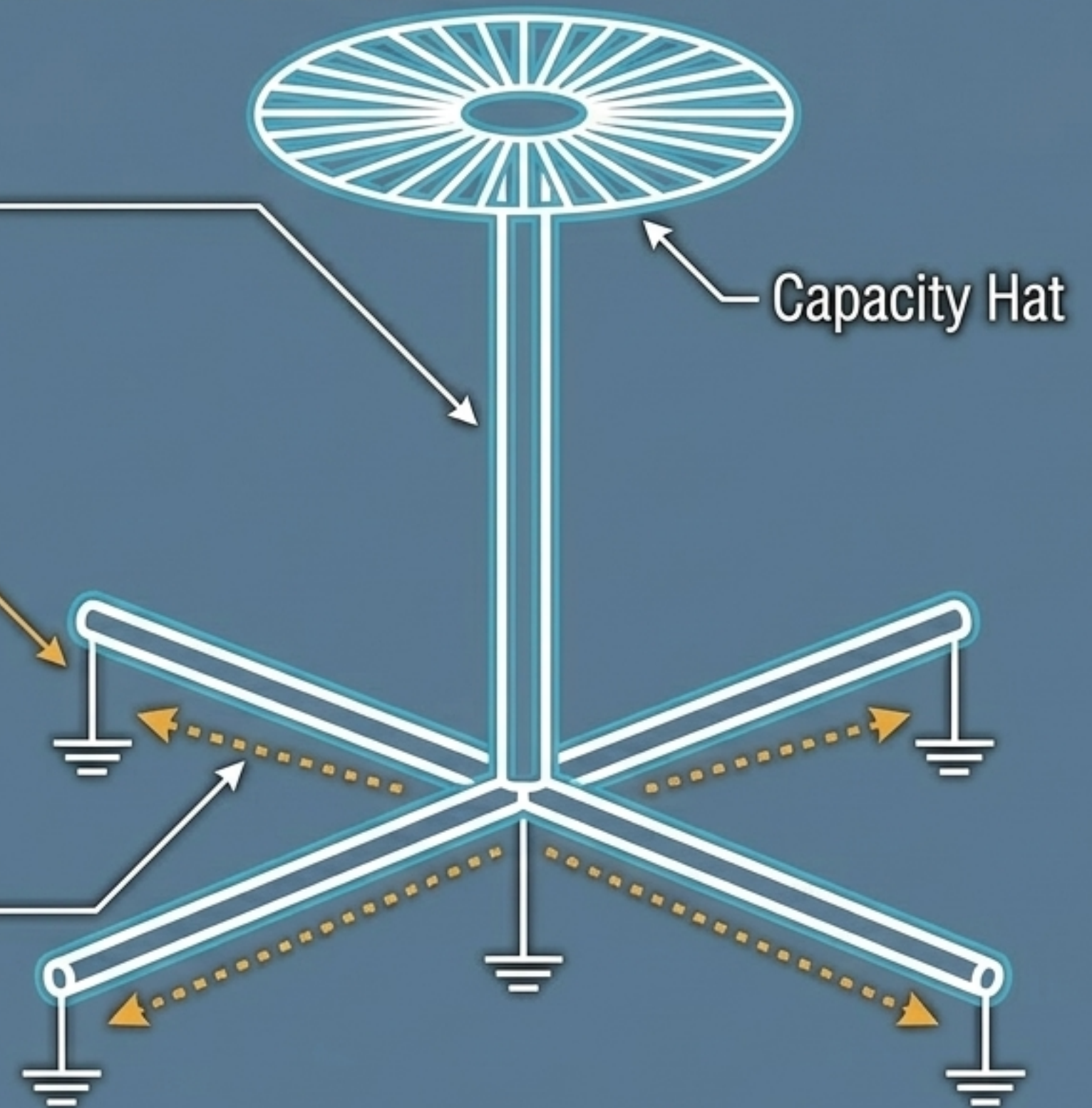
Omnidirectional horizontal pattern.
Provides excellent low-angle radiation
for DX (long distance).

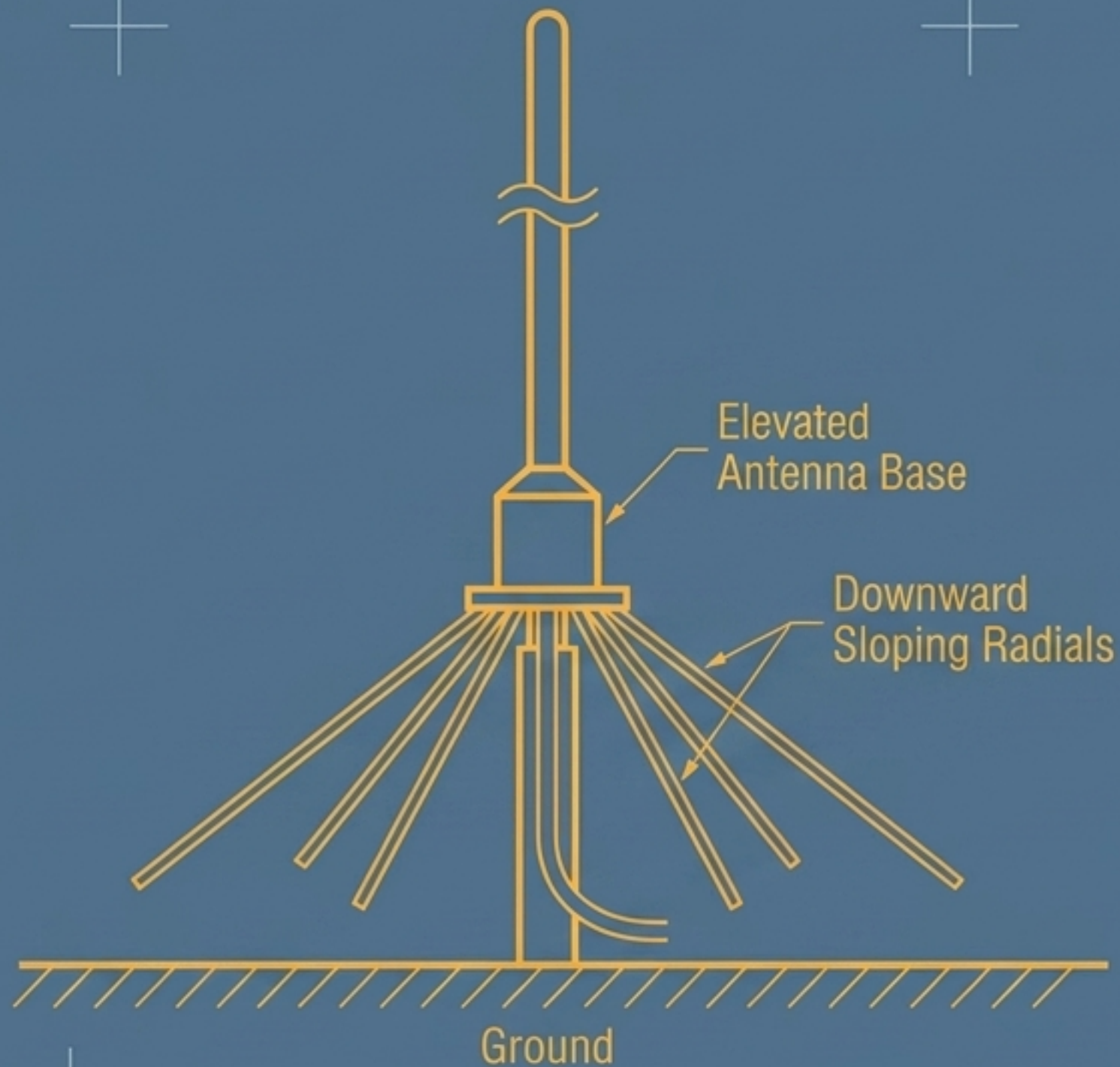
The Radial Effect •

Flat radials yield 36 ohm impedance.
Downward sloping radials artificially raise
impedance to exactly 50 ohms, perfectly
matching standard coax.

Capacity Hats & Loading Coils •

Components used to artificially compensate
for capacitive reactance on physically short
antennas at low frequencies.





Diagnostic Check: Vertical Systems

What configuration of radials will match an elevated quarter-wave vertical antenna to a 50-ohm coaxial cable?

- A) Downward sloping quarter-wave radials
- B) Horizontal quarter-wave radials
- C) Horizontal half-wave radials
- D) Upward sloping half-wave radials

Directional Arrays: Yagi-Uda & Quads

Reflector:

Placed at the back. The longest element (~5% longer). Inductively **reactive, lagging phase**.

Driven Element:

The center element. Connected directly to the coax line or balun.

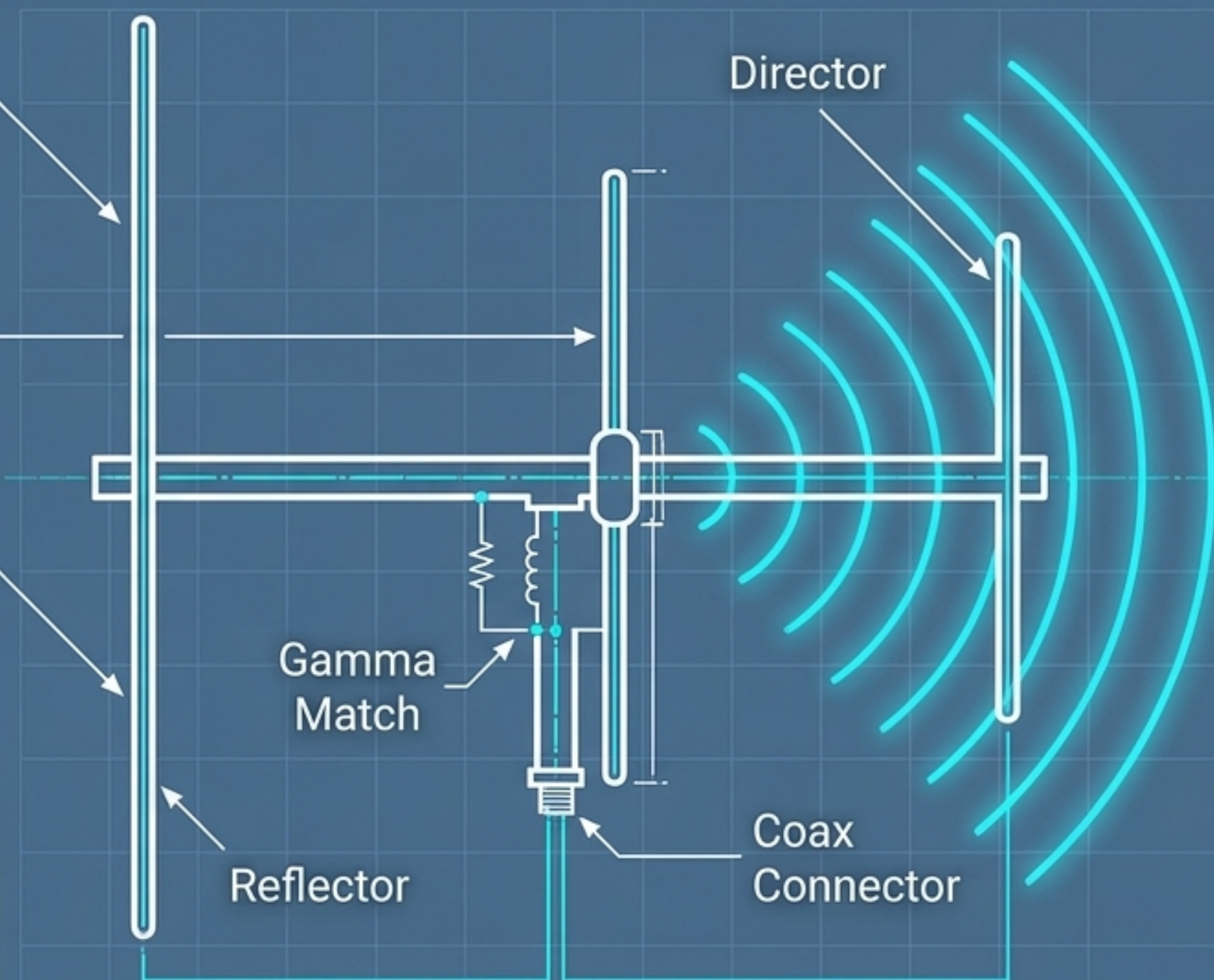
Director:

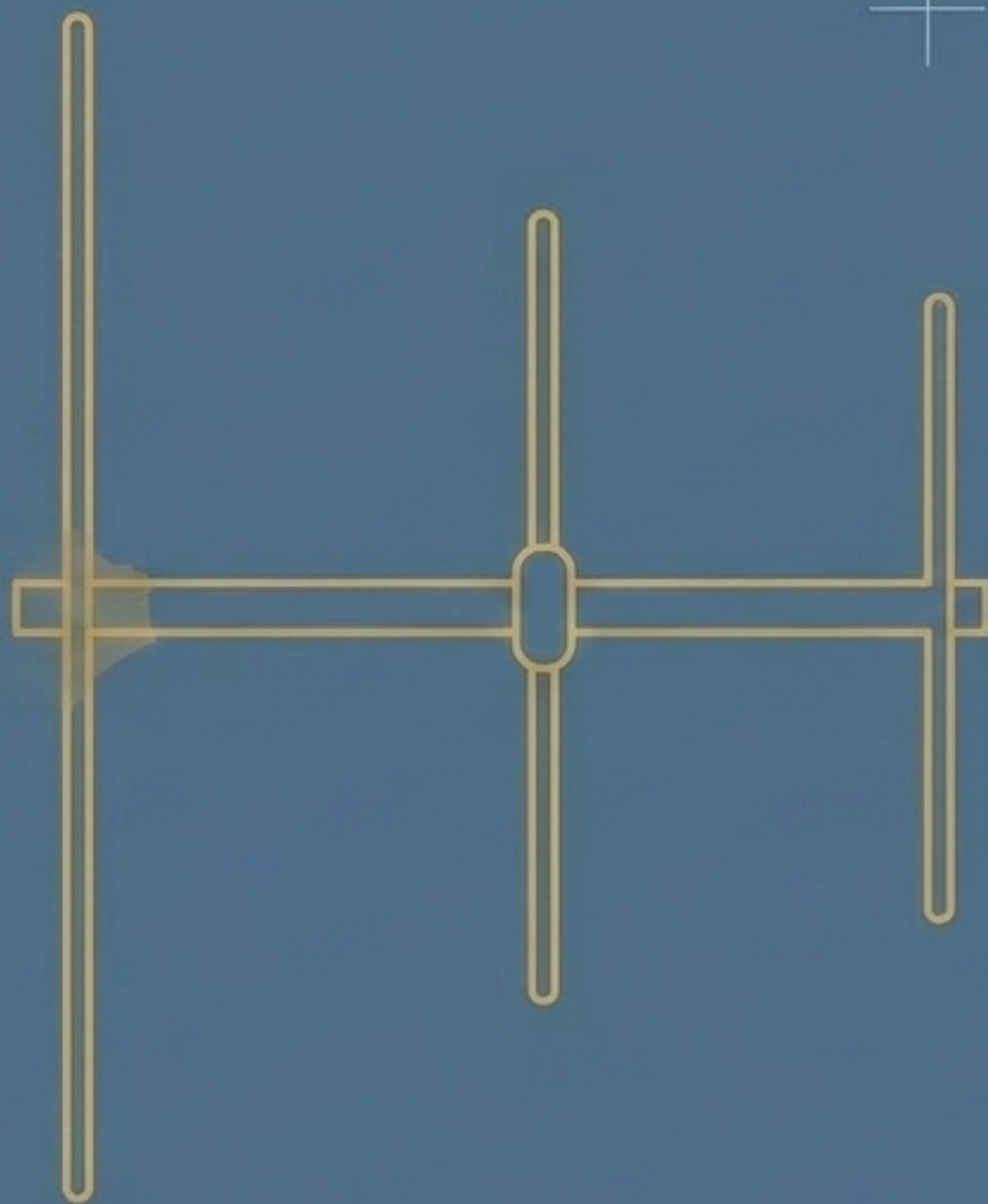
Placed at the front. The shortest element (~5% shorter). **Capacitively reactive, leading phase**.

Tutor's Toolkit

Quad Loop Variant:

Uses full-wave wire loops instead of straight rods.
Driven element length is calculated as $L = 306 / F(\text{MHz})$.





Diagnostic Check: Parasitic Antennas

In a 3-element Yagi antenna, what is the longest radiating element?

- A) The director
- B) The driven element
- C) The reflector
- D) The boom